Physics-Informed Attention Networks for Flow in Digital Porous Media

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Abstract. In the era of clean energy, flow dynamics in porous media have become pivotal for testing, designing, and developing efficient energy systems for environmental sustainability. Classical approaches, including physics-based numerical solvers, are effective in capturing these dynamics but are often computationally expensive and require extensive user intervention.In recent years, machine learning models such as recurrent neural networks (RNNs) and related architectures have shown promising results for accelerating flow simulations in digital porous media. Building on these advances, we propose a physics-informed attention-based deep convolutional recurrent neural network to simulate the underlying flow physics on digital rock images. The model leverages digital pore images (synthetic structures and actual micro-CT scans) as inputs, while simulation results obtained from the Lattice Boltzmann Method (LBM) serve as outputs for training, testing and validation. Our results demonstrate that the integration of a physics-driven customized loss function performs better than the normal loss function-based model does. The model has a comparative computational cost, enabling fast and reliable prediction of steady-state velocity fields at the pore scale. This approach offers a scalable pathway for efficient digital rock physics and clean-energy applications.

Keywords: Pore-scale Simulations · Lattice Boltzmann method · Physics informed neural net.

1 Introduction

The characterization of fluid flow behavior in porous media is a cornerstone of subsurface engineering, dictating the success of critical applications such as hydrocarbon recovery, geothermal energy extraction, groundwater remediation, and, more recently, carbon capture, utilization, and storage (CCUS) and underground hydrogen storage (UHS). Traditionally, the industry has relied on two

primary pillars for flow characterization: experimental core flooding and numerical simulation [1–5]. While laboratory experiments, such as steady-state permeability tests and micro-CT imaging, provide the most reliable representation of physical reality, they are expensive and time-consuming, often taking weeks to months to yield results for a single core sample. Numerical simulations offer a more practical alternative by mimicking operational changes and identifying optimized parameters, although they remain computationally demanding. Digital Rock Physics (DRP) has emerged as a complementary approach, utilizing high-resolution X-ray computed tomography (micro-CT) to digitize rock microstructures; on these digital samples, flow is simulated using classical Computational Fluid Dynamics (CFD) approaches, such as the Finite Volume Method (FVM) or the Lattice Boltzmann Method (LBM). The LBM, in particular, has become the standard for pore-scale modeling due to its ability to handle complex boundary conditions and multiphase interactions within the intricate geometry of porous media [6]. However, despite their accuracy, these physics-based numerical solvers face a significant bottleneck: computational cost. Resolving flow through the millions of voxels contained in a standard micro-CT scan requires massive computational resources and High-Performance Computing (HPC) clusters, rendering real-time sensitivity analysis and field-scale upscaling impractical.

To bridge the gap between computational efficiency and physical accuracy, deep learning models have recently been introduced as surrogate solvers for flow simulations. Early iterations of these models primarily utilized Convolutional Neural Networks (CNNs) to map pore geometry directly to velocity or permeability fields [7]. However, the challenge remains in developing models that are both computationally efficient and physically consistent, highlighting the need for approaches that integrate domain knowledge with data-driven techniques. Deep learning models are increasingly being developed for flow-related applications; for instance, [8] demonstrated the capability of 2D CNNs to predict permeability from binary images, while [9] extended this to 3D architectures (PoreFlow-Net), achieving significant speedups over direct numerical simulation. While these data-driven "black box" models demonstrated varying degrees of success, they often suffered from poor generalization when applied to rock types outside their training data. Furthermore, generative models like Generative Adversarial Networks (GANs) have been used to reconstruct digital rock structures [10], but they often struggle to preserve hydraulic connectivity without explicit physical constraints. Recognizing these limitations, the research focus has shifted toward Physics-Informed Machine Learning (PIML), which integrates domain knowledge and governing equations directly into the neural architecture or loss landscape.

[11] employed physics-constrained U-Nets to predict spatial and temporal variables, such as pressure and saturation, in multiphase flow through three-dimensional heterogeneous porous media; their approach captured the global nonsmoothness of pressure, achieving a 37% reduction in relative error compared with models without smoothing, while also lowering the mean absolute error by more than 4 psia. [12] demonstrated the effective use of physics-informed neu-

ral networks (PINNs) in conjunction with Welge's construction for multiphase flow, improving robustness in solving Buckley–Leverett-type equations. Furthermore, [13] proposed a hybrid model incorporating a novel physics-constrained neural network (PcNN) to ensure real-time consistency and enhanced generalizability in flow prediction. [14] embedded an analytical homogenization in a PINN for rapid permeability upscaling of digital rock, enabling accurate core-scale predictions on large 3D samples. [15] introduced a residual-based adaptive PINN that places collocation points where errors are largest to better track moving fronts in heterogeneous domains. [16] proposed a physics-informed graph neural network that treats the grid as a graph and embeds continuity and momentum to infer steady-state velocity fields across irregular pore structures. Although many surrogate approaches are demonstrated on 2D slices for efficiency, 2D formulations have inherent limitations: a single slice can lose true 3D connectivity by missing pore–throat links through the sample thickness, potentially misrepresenting percolation pathways, tortuosity, and connectivity even when the 2D geometry appears realistic; moreover, if training targets and constraints are restricted to in-plane velocities (U_x, U_y) with 2D physics, the model cannot represent out-of-plane flow (U_z) or enforce true 3D incompressibility; finally, slice selection can introduce bias (e.g., XY vs. XZ vs. YZ), so materials that look similar in 2D may still behave differently under 3D flow. Despite these caveats, 2D modeling is useful when the medium is effectively layered or when only planar observations (e.g., thin sections) are available, and it can provide a fast surrogate that tracks 3D trends without the cost of full 3D simulation; in our setup, augmenting the input with a pore mask, an EDT-like channel, and a small "drive" vector helps the network learn morphology-to-flow relationships at substantially lower computational cost. True 3D models are considerably more memory- and time-intensive because volume data scale cubically, and training will likely require smaller 3D patches, fewer channels, more aggressive mixed precision, and/or gradient checkpointing compared to the current 96×96 2D setup. Collectively, these studies underscore the emerging trend of integrating domain knowledge and governing equations into neural architectures; therefore, this study implements a customized loss function to simulate flow behavior in porous media and compares the results with those of a traditional loss function.

2 Methodology

We deployed a physics-informed 2D Convolutional Recurrent Neural Network (ADCRNN2D) to predict steady-state velocity fields in porous media from pore-space images. The proposed model combines CNN encoders, ConvLSTM recurrence, deformable attention, a decoder, and a customized loss function, as is shown in Figure 1.

The input tensor includes a raw digital porous medium comprising of a voxelized pore structure along with boundary/initial conditions. This digital analog of the rock microstructure represents the pore-geometry and flow-driving conditions. The input sample considered has the shape [4,3,96,96] with the spatial

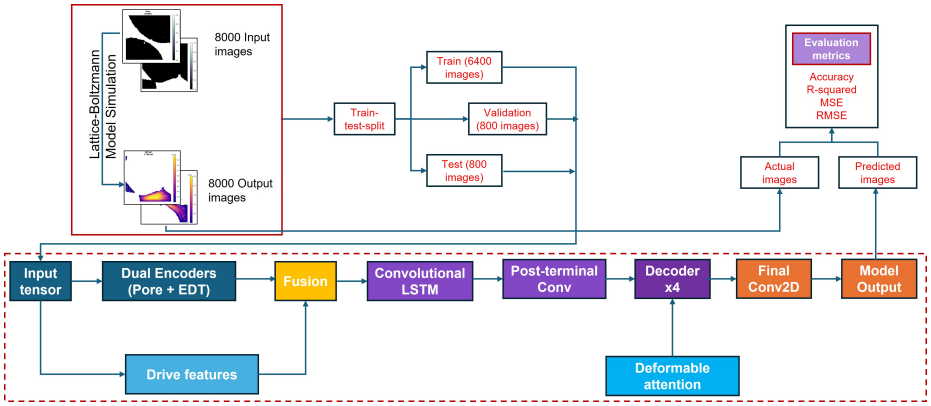

Fig. 1: Proposed framework for this study.

dimensions of 96×96, 24 batch sizes and three input channels corresponding to one binary pore mask and two tiled components of the applied driving force. The two parallel encoders included pore structure and Euclidean distance transform (EDT), which extracts structural constraints (pore connectivity, throat size, and tortuosity) and distance-to-solid information (hydraulic radius and permeability pathways). The pore structure encoder and EDT change the representation of the original image to [4,768,4,4] for both encoders. The resulting encoders are concatenated along the channel dimension. The drive features induce parameters such as the pressure gradient, which determines the behavior of the flow. The fusion of encoders and drive features is used to merge the structural and driving information, which corresponds to combining the porous structure with applied forces. The Convolutional long short-term memory captures the fluid memory effects along with the sequential propagation of velocity through connected pores. The post terminal convolution enforces the local consistency of velocity gradients. 4 Decoders translates the learning from the coarse-scale flow patterns back into fine-scale velocity fields. Deformable attention allows queries to be considered for a compact set of semantically relevant positions inside the pore images and thereby improves the computational efficiency and geometrical flexibility. The final Conv2D converts latent features into flow behavior parameters such as the velocity field in the form of two channels representing the normalized velocity components in the horizontal and vertical directions.

The velocity components in the pore spaces are given by the following equation:

$$\hat{U}\,(x,y) = (U_x(\,x,y), U_y(x,y)), \hat{U}\epsilon \mathrm{R}^{[4,2,96,96]} \tag{1}$$

The main hypothesis of this study was to implement the ADCRNN2D model with different loss formulations to evaluate steady-state velocity predictions on the Ketton dataset. We compared a masked mean squared error (MSE) loss and

a customized physics-informed loss. The masked MSE loss is defined as

$$L_{MSE} = \frac{1}{2\sum_{x,y} M(x,y)} \sum_{x,y} M(x,y) \parallel \hat{v}(x,y) - v(x,y) \parallel^2, \tag{2}$$

where $M(x,y)$ restricts error evaluation to the pore space.

The customized physics-based loss (L_{custom}) combines vector MSE (L_{vec}), Charbonnier loss (L_c), projection loss (L_{proj}), divergence penalty (L_{div}), and gradient regularization (L_{grad}):

$$L_{custom} = L_{MSE} + \lambda\big(0.55\, L_{vec} + 0.25\, L_c + 0.15\, L_{proj} + 0.05\, L_{div}\big) + 0.08\, L_{grad}. \tag{3}$$

Gradient regularization penalizes mismatches in spatial derivatives of the predicted vs. target velocity fields (within the pore mask), helping preserve local flow structure; its weight (0.08) was tuned empirically to strengthen early learning without destabilizing training.

All other coefficients were also chosen empirically to balance accuracy, stability, and physics: 0.55 L_{vec} prioritizes pointwise velocity fidelity, 0.25 L_c adds robust stabilization against localized outliers, and 0.15 L_{proj} encourages alignment with the driving-force direction without dominating reconstruction terms. The divergence penalty is given a smaller weight (0.05 L_{div}), yet it serves as the primary incompressibility constraint, significantly enhancing the physical plausibility of the predicted flow fields. Additionally, the overall physics-loss weight λ was gradually increased 0.03 to 0.15 during training using a cosine schedule, allowing the network to first learn basic velocity patterns before progressively enforcing stronger physical constraints.

2.1 Dataset description

We begin with a high-resolution X-ray micro-CT scan of four different carbonate and sandstone rocks (Bentheimer, Doddington, Estaillades, and Ketton), which is stored as a 3D raw image with dimensions $N \times N \times N$, where $N = 1000$. A subvolume of size $200 \times 200 \times 200$ voxels was extracted from the original dataset. Image segmentation was performed using intensity thresholding. Voxels with grayscale values below the chosen threshold were labeled as pore space, while the remaining voxels were considered as the solid matrix. To build a diverse and realistic dataset, the segmented binary volume was divided into overlapping 2D patches of size 96×96 with a stride of 48. Patch selection was stratified by porosity and uniformly sampled across 16 porosity bins ranging from 15 to 30%, with approximately 133 samples per bin for each rock type. This ensured balanced representation across the porous media and prevented over-representation at any single porosity level. Each patch contained a sufficient fraction of pore space to ensure non-trivial flow.

We simulated fluid flow inside each pore patch using the Lattice Boltzmann Method (LBM), which models two-dimensional fluid flow using discrete distribution functions with nine velocity directions. The LBM update equation incorporates both the relaxation rate and the equilibrium distribution, along with an external body force. Boundary conditions enforce the no-slip constraint at solid surfaces through the bounce-back scheme.

During the LBM simulations, periodic boundary conditions were applied along all domain edges using circular streaming. No-slip walls were enforced at solid–pore interfaces via bounce-back boundary conditions after the streaming step. Additionally, velocity components within solid regions were explicitly masked and set to zero after the collision step to ensure physical consistency. For each valid simulation, the pore mask, applied driving force vector, and resulting velocity field components were stored in Hierarchical Data Format 5 (HDF5). After excluding invalid simulations exhibiting zero flow, the final dataset comprised 8000 valid pore-scale samples. Furthermore, pore-scale images were restricted to porosity values between 10 and 30% to reflect realistic field conditions. These samples were subsequently used for training, validation, and testing of the proposed model.

3 Results and discussion

3.1 Experimental setup

We partitioned the dataset into training (80%), validation (10%), and testing (10%) subsets to evaluate generalizability. The model was trained for 150 epochs with batch size 24 with gradient accumulation over two steps to meet memory constraints while maintaining stable optimization. Training used AdamW with linear warmup (first 10 epochs) followed by cosine annealing warm restarts, with a maximum learning rate of 2.4×10^{-4} and weight decay 2×10^{-4}. Early stopping (patience: 100 epochs) and adaptive learning-rate decay were applied: if validation loss did not improve for 10 consecutive epochs, the learning rate was halved down to a minimum of 1×10^{-7}, and the best model was selected by minimum validation loss. Gradient clipping (max norm 0.7), mixed-precision training, and cuDNN auto-tuning were enabled for stability and GPU efficiency. During training, a binary pore mask was provided as an input channel and MSE, gradient, and divergence losses were evaluated only in the pore space by masking solids via element-wise multiplication with $1 - \text{pore_mask}$, thereby supervising zero-velocity in solid regions. Whereas LBM enforces no-slip via bounce-back, the network learns interface velocities implicitly through masked supervision; future work may add explicit interface constraints via penalty terms or architectural changes. Training and validation losses tracked for 150 epochs (Loss Tracking figure), indicating robust generalization without overfitting. Regularization further included spatial dropout (15% in attention/LSTM, 7.5% in the decoder), complementing weight decay and gradient clipping to maintain stability across the 150-epoch run.

3.2 Comparative analysis

The customized loss function enhances the ability to capture the flow dynamics in digital porous media. First, we fixed 150 as the number of epochs for the comparison of the two models based on model accuracy, as depicted in Figure 2. We then evaluated the performance of these two models using evaluation metrics, including R-squared, mean square error, root mean square error and structural similarity index measure. The incorporation of the customized loss function outperforms the model based on the traditional loss function, as shown in Table 1 and Table 2. Results suggest that implementing the customized loss function improves the performance of deformable-attention models for digital rock processing.

Table 1: Best performance metrics during evaluation of the training dataset using deep-learning architecture for the said task

Methods	R-squared	MSE	RMSE	SSIM
ADCRNN with masked MSE loss function	0.718	1.15	1.043	0.7952
ADCRNN with customized loss function	0.838	0.825	0.91	0.85

Table 2: Best performance metrics during evaluation of the validation dataset using deep-learning architecture for the said task

Methods	R-squared	MSE	RMSE	SSIM
ADCRNN with masked MSE loss function	0.631	1.404	1.140	0.7909
ADCRNN with customized loss function	0.768	0.954	0.983	0.8251

Typical outputs for both architectures along with their heatmaps are presented in Figure 3 and Figure 4. The highlighted regions illustrate the velocity fields for the input geometry and the corresponding predictions produced by the two deep-learning architectures. The observed velocity range indicates that the customized loss function improves the predictive capability of the deformable-attention model. In terms of computational cost, the total training time for the MSE-loss model was 4h 9m 55s (14995.03s), while the customized-loss model required 4h 10m 51s (15051.20s). However, the prediction time for both loss formulations is approximately 10 seconds for 800 samples. For the LBM simulations, using GPU CuPy required around 2 min 30 s to generate 2000 velocity samples per rock type; thus, while model training is comparatively time-consuming, the trained ML surrogate enables substantially faster sample generation during inference.

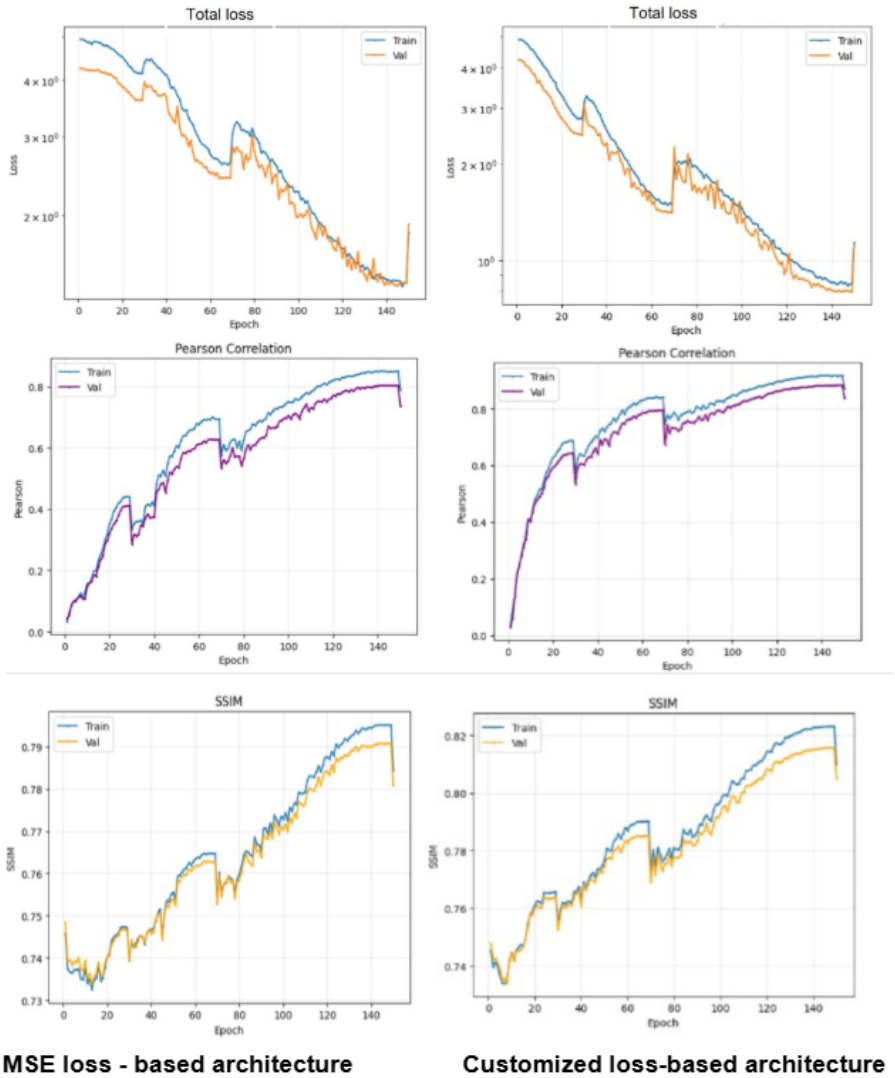

Fig. 2: Total loss and Pearson correlations for validation and training with respect to epochs for (a) ADCRNN with conventional loss function and (b) ADCRNN with customized loss function.

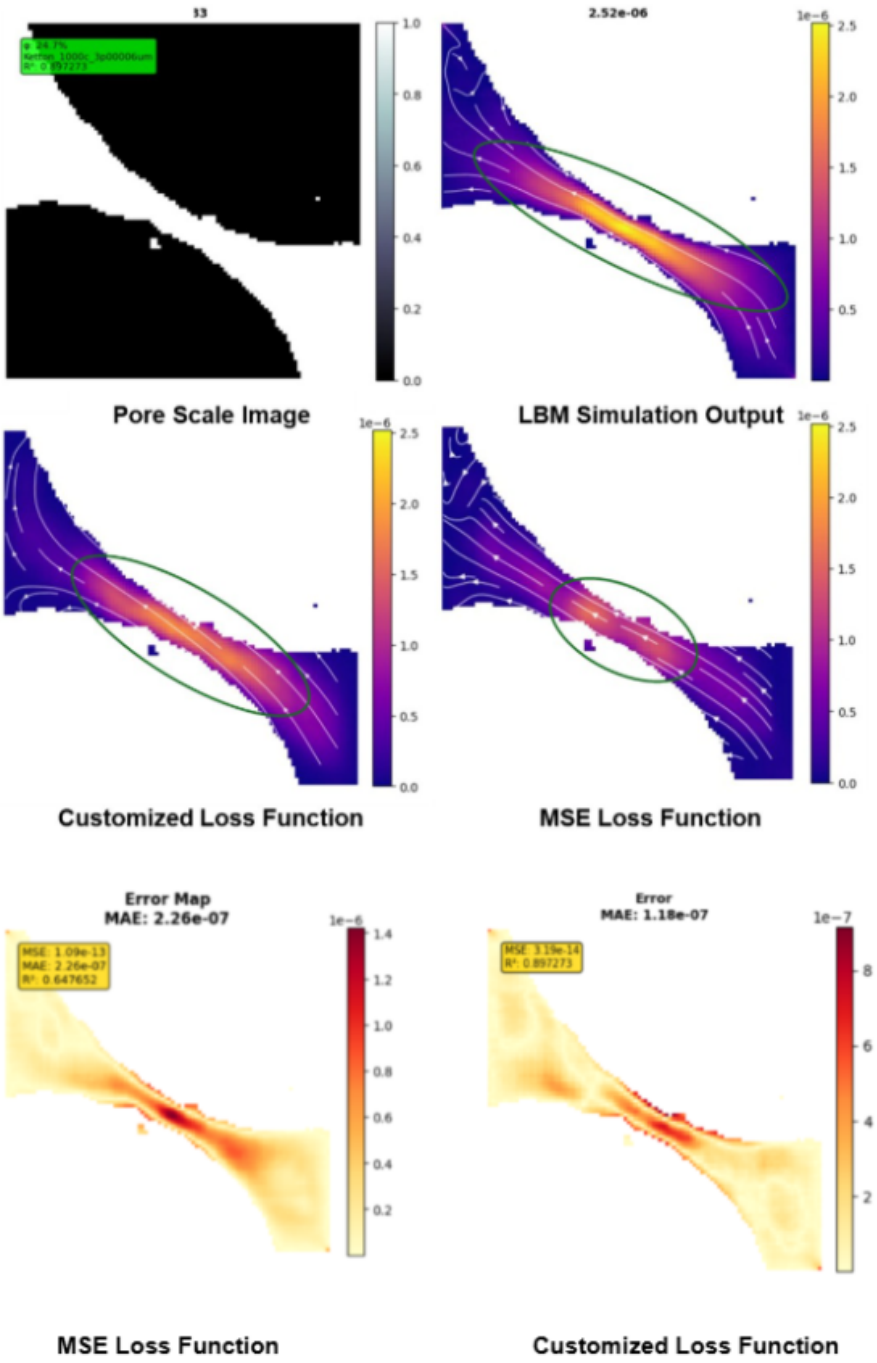

Fig. 3: Sample model output comparison with respect to Lattice Boltzmann model simulation and (a) ADCRNN with conventional loss function and (b) ADCRNN with customized loss function.

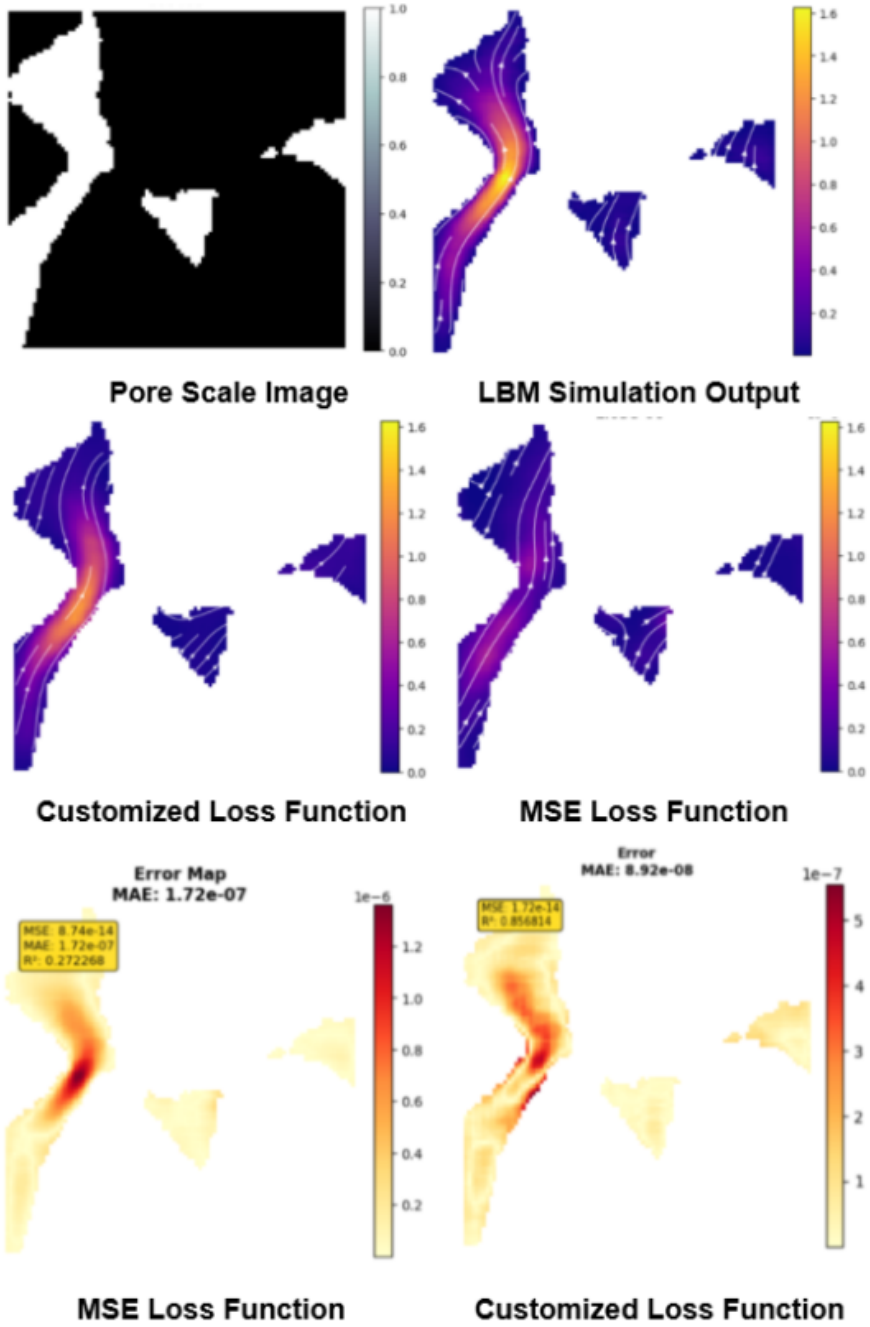

Fig. 4: Another sample model output comparison with respect to Lattice Boltzmann model simulation and (a) ADCRNN with conventional loss function and (b) ADCRNN with customized loss function.

4 Conclusions

In this study, we addressed the computational inefficiency of traditional pore-scale flow simulations by developing and validating a Physics-Informed Attention-based Deep Convolutional Recurrent Neural Network (ADCRNN) to predict steady-state velocity fields in digital porous media at deep-learning inference speed while preserving the accuracy of numerical solvers such as the Lattice Boltzmann Method (LBM). The core innovation is an architecture that integrates Convolutional LSTM units with Deformable Attention, coupled to a physics-driven customized loss designed to go beyond pixel-wise fidelity by enforcing vector alignment, divergence (mass conservation), and gradient consistency, thereby encouraging adherence to the Navier–Stokes physics governing transport. The advantages of this physics-informed formulation are demonstrated by the following pore-scale simulation samples: the customized-loss ADCRNN achieved $R^2 = 0.838$ (training) and $R^2 = 0.768$ (validation), while a standard masked MSE baseline reached only $R^2 = 0.718$ (training) and $R^2 = 0.631$ (validation); consistent with this, the customized model reduced the validation RMSE to 0.983 compared to 1.140 for the baseline. Qualitative velocity-field visualizations further confirm that the customized loss better preserves high-velocity channels and complex throat connectivity that are often over-smoothed by traditional losses, yielding close agreement between LBM ground truth and ADCRNN predictions and indicating that the model learns the nonlinear mapping from pore geometry to fluid momentum. The model also demonstrates strong generalization within the 15–30% porosity training range across four diverse rock types, suggesting practical reliability in this porosity regime, with the physics-informed loss providing $\sim$ 20% accuracy gains over standard MSE.However, deployment to extreme porosities ($< 15\%$) should be validated carefully because pore connectivity and flow regimes can change qualitatively outside the training distribution. Beyond velocity prediction, this framework offers a scalable alternative to LBM for large-scale digital rock physics workflows by substantially reducing flow-characterization time, enabling rapid sensitivity analysis and permeability upscaling relevant to carbon sequestration and hydrogen storage. Future work will (i) expand the dataset to additional lithologies, including complex carbonates with microporosity, to further test generalizability, (ii) extend the current 2D patch-based approach to full 3D volumes to capture true rock topology, and (iii) adapt the physics-informed loss to multiphase flow by incorporating saturation and capillary-pressure constraints toward a comprehensive real-time reservoir simulation tool.

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